SPE-226650-MS

Experimental Evaluation of Fracture Conductivity for Acid-Fracturing of Ultra-High Temperature Carbonate Reservoir

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This paper was prepared for presentation at the SPE International Hydraulic Fracturing Technology Conference and Exhibition held in Muscat, Sultanate of Oman, 23 - 25 September 2025.

Summary

Acid-fracture conductivity plays a critical role in enhancing oil and gas production from carbonate reservoirs, and laboratory experiments remain the primary approach for its investigation. However, recent discoveries of carbonate reservoirs in China, characterized by depths exceeding 10,000 meters and temperatures ranging from 160°C to 220°C, present significant challenges for current experimental methodologies. The existing testing equipment, with a temperature resistance limit of 160°C, is insufficient to accurately simulate acid etching behavior under such high-temperature conditions.

Therefore, a novel ultra-high-temperature API acid-etching system was developed, capable of withstanding up to 240 °C and operating stably at 220 °C, which employed a three-stage heating process involving the acid-tank, pipelines and rock slabs holder. Additionally, acid was circulated within a fully sealed system to capture CO_2 and Ca^{2+}, Mg^{2+} produced during the acid-rock reaction in the fracture. This approach facilitates a more accurate simulation of in-situ acidizing conditions. The acid-fracture morphology was characterized by a 3D laser scanner, and fracture conductivity was quantitatively evaluated.

The research results show that the acid-rock reaction rate increases initially exponentially with temperature, then linearly; at 180 °C, it reaches 11.45 times the rate at 90 °C. The "peak-removing" effect of acid-etching on fracture becomes more pronounced with increasing temperature. At 160 °C and above, acid uniformly dissolves surface protrusions, widening and deepening valleys, and forming a uniform single-sided channel at 180°C. However, below 160 °C, acid dissolves protrusions locally, only widening valleys. Under closure pressure, conductivity declines transitioned from linear to exponential with increasing temperature. At 80 MPa, the acid-fracture conductivity at 180°C is only 67% of that observed at 90 °C.

This study provides a robust experimental device and methodology for assessing acid-fracture conductivity in ultra-high temperature carbonate reservoirs. And future advancements in acid systems capable of withstanding temperatures above 200 °C are expected to provide deeper insights into acid-etching behavior and fracture conductivity under such extreme conditions.

Introduction

With the continuous advancement of oil and gas exploration technologies, high-quality carbonate rock reservoirs have been discovered in ultra-deep reservoir (exceeding 7,000 m) in both the Sichuan and Tarim basins of China (Ma et al., 2024). The temperature of such reservoirs typically ranges from 160°C to 200°C (Gao et al., 2024; Gou et al., 2024), with the TK1 well in the Tarim Basin reaching a completion depth of 10,910 m and the temperature of 210°C. Acid fracturing is a key stimulation technology for production in carbonate reservoirs, as it employs acid to non-uniformly dissolve hydraulic fractures under reservoir temperature, forming acid-etched fractures with high conductivity, thereby providing efficient flow channels for hydrocarbons (Kong et al., 2024; Su et al., 2023). The length and conductivity of these acid-fractures are critical factors that determine the performance of acid fracturing (Kong et al., 2023; Pournik et al., 2009). However, the extreme conditions in ultra-deep, ultra-high-temperature carbonate reservoirs present significant challenges to traditional acid fracturing techniques. On one hand, the great reservoir temperatures accelerate the acid-rock reaction rate, leading to rapid acid consumption near the fracture entrance, which limits effective etching at greater distances (Fan et al., 2023; Guo et al., 2020). For instance, when the temperature increases from 80°C to 149°C, the acid-rock reaction rate of 28% HCl rises by 2.6 times (Sayed et al., 2023). On the other hand, under the high closure pressure of ultra-deep reservoirs, the rough surfaces of acid-fractures are prone to crushing, resulting in a reduction in fracture width and conductivity (Liu et al., 2024; Zhang et al., 2018). This dual challenge of "low efficiency of acid" and "rapid decline of fracture conductivity" significantly diminishes the performance of acid-fracturing in ultra-deep and ultra-high-temperature reservoir.

Laboratory experiment is the most direct method for investigating the acid-fracture conductivity (Chen et al., 2025; Cui et al., 2024). However, experimental studies on acid-fractures at temperatures above 160 °C are rarely reported (Table. 1). Navarrete conducted acid-etching experiments at 160 °C and demonstrated that 20% HCl tended to excessively dissolve the fracture entrance, whereas retarded acid systems achieve a more uniform etching along the fracture surface (Navarrete et al., 2000). Nevertheless, their device is not suitable for higher-temperature conditions.

Table 1—Acid-etching equipment and working parameters.(Gou et al., 2025)

Device	Operating temperature (°C)	Core geometry	Pumping rate (ml/min)	Fluid type	Author
Plunger core	26.7	5 cm×15 cm	/	Blank HCl, formic acid, acetic acid	Smith et al., 1970
	80	10 cm × 40 cm	4.32-64.8	Gelled acid, organic acid, emulsifying acid	Ruffet et al., 1998
Rock sheet acid etching	38-93.	2.54cm thick, 6.35-8.89cm diameter	67.5	Blank HCl, retarded acid	Broaddus et al., 1968
	93	1.27cm thick, 5.7-10.2cm diameter	/	Foamed, gelled, emulsified, foamed, organic acid	Anderson and Fredrickson, 1989
Rock slab acid etching device	/	5.1cm × 7.6cm × 2.5cm	1.4-2.0	Blank HCl	Beg et al., 1998; Malik and Hill, 1989
	160	API	1000	Blank HCl, emulsified acid	Navarrete et al., 2000
	65	API	1000	Gelled acid	Suleimenova et al., 2016
	80	100cm×25cm	20-30	Blank HCl, gelled acid and Jelly acid	Li et al., 2017
	130	API	50	Gelled acid	Zhang et al., 2018
	93	API	1040	Blank HCl, gelled acid	Malagon et al., 2006

Device	Operating temperature (°C)	Core geometry	Pumping rate (ml/min)	Fluid type	Author
	110	100 cm × 10 cm × 1.5 cm	1200	Blank HCl	Liu et al., 2021
	120	API	2	Autogenic acid	Zhang et al., 2023
	140	2.5×7.5cm, splice length 3 m	6420	Blank HCl	Roberts, 1974

In summary, research on acid-fracture morphology and conductivity under ultra-high temperature conditions (≥180 °C) remain limited, primarily due to temperature constraints of existing experimental equipment which restrict the simulation of acid-etching behavior in hydraulic fractures. To address this limitation, this study developed an API-standard acid-etching experimental device capable of withstanding up to 240 °C and operating stably at 220 °C. Acid-etching experiments were conducted under ultra-high temperature, and acid-fracture morphologies were quantitatively characterized by a 3D laser scanner. Additionally, fracture conductivity tests were performed to examine conductivity evolution. The results provide enhanced insight into acid-fracture behavior in ultra-high-temperature carbonate reservoirs and support the design of acid fracturing treatments in such environments.

Experimental Method

Experimental Samples

In the PS gas field of the Sichuan Basin, the DY Formation dolomite reservoirs are buried at depths exceeding 6,500 m, with an average temperature of 180 °C and closure pressure of 80 MPa. To study acid-etching behavior in such ultra-deep, high-temperature conditions, experiments were conducted using DY Formation outcrop samples. The outcrop is predominantly composed of dolomite (about 98.7%), with small amounts of plagioclase and clay minerals (Fig. 1). The outcrop samples were cut into API-standard rock slabs (178 mm×38 mm×50 mm). To simulate acid-etching behavior in the reservoir, rough fracture surfaces were created using the Brazilian splitting method (Fig. 2). Fracture roughness was characterized using the JRC parameter (Wang et al., 2023), and five samples with similar JRC values were selected to reduce the impact of initial surface variations (Fig. 3).

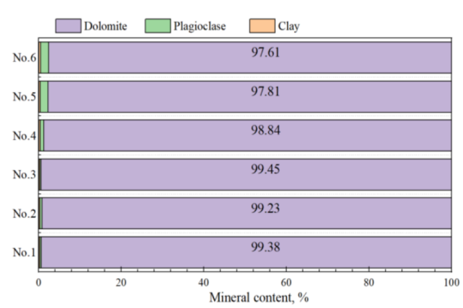

Figure 1—Mineral composition of the outcrop samples

Figure 2—Rough surface of the rock sample

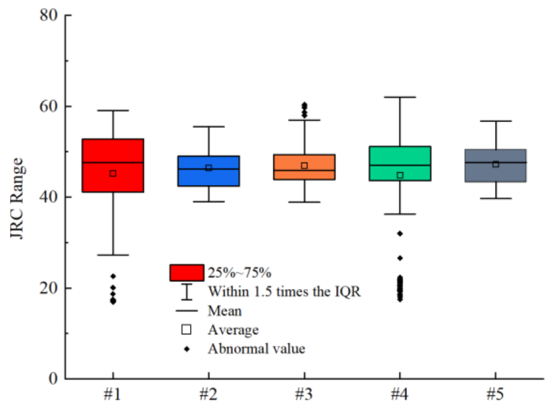

Figure 3—JRC of the rough fracture surface before acid-etching

Gelled acid was used in this study, consisting of 20.0% HCl, 1.5% gelling agent, 5.5% corrosion inhibitor, 1.0% iron stabilizer, 2.5% corrosion inhibitor intensifier.

Experimental Parameter and Scheme

A series of acid-etching experiments were conducted at temperatures ranging from 90 °C to 180 °C to investigate the etching behavior and conductivity evolution of acid-fracture. The laboratory acid injection rate was determined based on field-scale pumping rate using the Reynolds number similarity criterion (Eq. 1) (Gou et al., 2021),

$$q_1 = \frac{q_f h_1}{2h_f}\left(\frac{w_f}{w_1}\right)^{\frac{2n-2}{2-n}} \tag{1}$$

where, w_f is fracture-width in the reservoir, m; q_f is field-scale pumping rate, m^3/s; h_f is fracture height in the reservoir, m; w_l is fracture-width between rock slabs, m; q_l is laboratory-scale pumping rate, m^3/s; h_l is width of the rock slabs, m; n is flow behavior index, dimensionless. Reference data were obtained from actual field operations (Table. 2), with an average acid injection rate of approximately 6 m^3/min. Based on this, the equivalent laboratory-scale rate was calculated as 150 mL/min. Experimental conditions for acid-etching tests are summarized in Table. 3.

Table 2—Calculation parameters for laboratory acid injection rate

q_l, mL/min	q_f, m^3/min	h_l, m	w_f, m	h_f, m	n	w_l, m
150	6.0	0.038	0.02	100	0.36	0.002

Table 3—Ultra-high-temperature fracture acid etching experimental scheme

No.	Temperature, °C	Acid pumping rate, ml/min	Time, min
#1	90		
#2	120		
#3	140	150	40
#4	160		
#5	180		

Ultra-High-Temperature Acid-Etching Experimental System

The acid-fracture conductivity experimental system (Fig. 4) includes an ultra-high-temperature acid-etching fracture device (Fig. 4a), a 3D laser scanner (Fig. 4b), and a fracture conductivity tester (Fig. 4c). The core component is a self-developed ultra-high-temperature acid-etching device designed by Southwest Petroleum University (Fig. 4a). It mainly consists of an acid storage system, a digital monitoring and control unit, an acid injection system, and a three-stage heating system.

a. Acid-etching device b. 3D laser scanning device c. Fracture conductivity testing device

Figure 4—Ultra-high-temperature acid-etching fracture conductivity system

Traditional acid-etching devices face challenges in rapidly heating the acid to 180 °C and maintaining a reliable seal at temperatures exceeding 180 °C. Furthermore, under ultra-high-temperature conditions, acid evaporation poses safety risks when using conventional open acid pumping methods. To address these issues, a novel acid-etching holder was developed (Fig. 5). This holder employs solid components to securely fixes rock slabs on both sides, with a central acid flow channel integrated to form the rock slab holder. This design ensures that the acid reacts exclusively with the fracture surface and maintains a tight seal even at 240 °C. Additionally, the system incorporates a three-stage heating system comprising the acid tank, pipelines, and the holder, combined with a fully enclosed acid pumping loop system (Fig. 6). This configuration enables rapid heating of the acid to 240 °C while effectively preventing acid vapor from escaping into the environment.

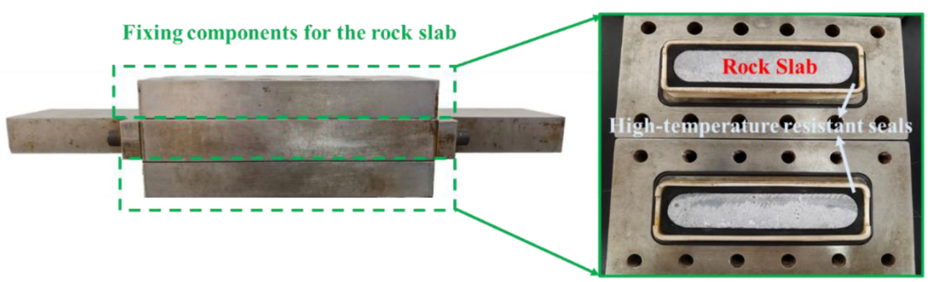

Figure 5—Rock slabs holder

Fig. 6 presents the schematic diagram of the ultra-high-temperature acid-etching fracture device. The entire system is driven by a water pump. An external container delivers acid into the upper compartments of Tanks 1 and 2. At the beginning of acid-etching experiment, fresh water is pumped into the lower compartments of Tanks 1 and 2, which displaces the acid from the upper compartments sequentially into the holder and subsequently into Tanks 3 and 4. Meanwhile, the fresh water in the lower parts of Tanks 3 and 4 is discharged through backpressure valves. Once the acid in Tanks 1 and 2 has been fully displaced, the next pumping cycle initiates. Fresh water is then pumped into the lower compartments of Tanks 3 and 4, displacing the acid from the upper compartments into the holder and then into the upper compartments of Tanks 1 and 2. Throughout this process, both the acid tanks and valve positions are switched to ensure that the acid is consistently injected into the holder from the same direction. This circulation continues until the completion of the experiment. During the entire process, Detector 1 and Detector 2 continuously monitor the temperature and pressure at the fracture inlet and outlet, respectively. The acid injection rate is precisely controlled by the water pump.

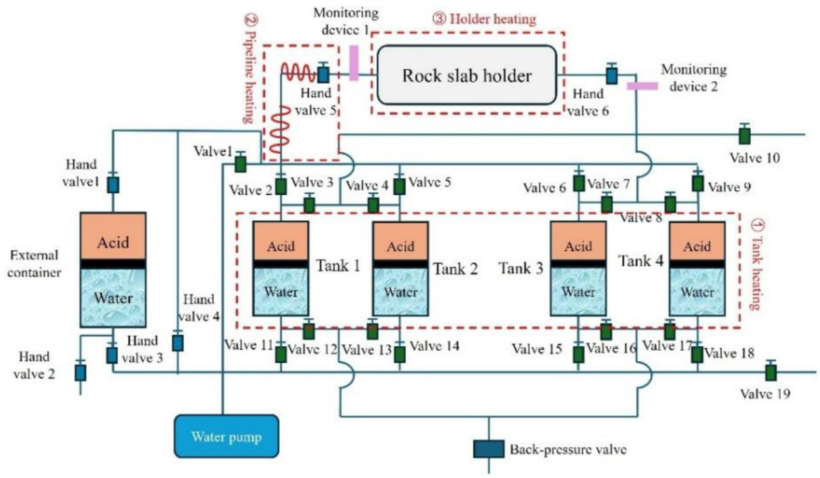

Figure 6—Schematic of the ultra-high-temperature acid-etching fracture device

As an example, the acid-etching fracture experimental procedure at 180 °C is described as follows:

- The rock slabs are placed into the holder and connected to the pipeline system. Subsequently, the sealing integrity of the entire setup is verified by pumping saline water through the system using a water pump.
- Once the system is confirmed to be fully sealed, 2 L of acid is injected into the acid tanks via an external piston container.
- The back pressure is set to 7.5 MPa, while the temperature of acid tank is maintained at 80 °C. The two-stage pipeline heating system is configured to 120 °C and 160 °C, respectively, and the holder temperature is set to 180 °C.
- After all sections have reached their designated temperatures, the acid is pumped into the holder. Data recording begins once the backpressure stabilizes at 7.5 MPa and continues until the experiment concludes.
- Upon completion of the experiment, the acid is drained from the system, and water is introduced to clean and cool down the equipment.

Following the acid-etching experiment, the rock slabs are carefully removed from the holder. Fracture elevation data is then acquired using a 3D laser scanner (Fig. 4b), and the dry weight of the rock slabs is measured. Finally, the rock slabs are installed in the fracture conductivity testing device (Fig. 4c) to evaluate the conductivity of the acid-fracture.

Experimental Results and Discussion

Dynamic Fracture Width during Acid-Etching

The dynamic acid-fracture width was determined using Eq. 2, based on the pressure differential between the fracture inlet and outlet and the acid pumping rate. The calculation was conducted based on Darcy's law (Zou et al., 2023) in conjunction with the fracture width calculation formula (Ruffet et al., 1998),

$$b = \left(\frac{12 \mu L q}{h \Delta p} \right)^{\frac{1}{3}} \tag{2}$$

where, b is the dynamic acid-fracture width, m; q is the experimental pumping rate (m^3/s); h is the width of the rock slab, m; μ is the viscosity of the acid, mPa·s; l is the length of the rock slab, m; Δp is the pressure differential between the fracture inlet and outlet, MPa.

As shown in Fig. 7, the pressure differential decreases exponentially within the temperature range of 90–140 °C. However, the etched fracture width exhibits different trends: at 90 °C, it increases linearly, while at 120 °C and 140 °C, it increases logarithmically (Fig. 7a, b, c). This indicates that at 90 °C, the rate of fracture widening and morphological evolution during acid-etching is relatively slow. At 120 °C and 140 °C, the initially narrow fractures generate a higher-pressure differential due to more intense acid-rock reaction. Over time, as the fracture widens, the area-volume ratio decreases, leading to a lower acid consumption and a slower rate of fracture propagation. At 160 °C, the pressure differential decreases logarithmically, and the fracture width increases in a comparable manner (Fig. 7d). This suggests that although the fracture continues to expand, the area-volume ratio has a minimal effect on the acid-rock reaction rate at this temperature. In contrast, at 180 °C, a significant pressure drop occurs within the first 13 minutes, followed by a relatively stable pressure trend (Fig. 7e). This phenomenon may be attributed to the rapid reaction rate at this temperature, leading to the early formation of flow channels or wider fractures, which subsequently stabilize the pressure. As illustrated in Fig. 7f, when the temperature exceeds 90 °C, the fracture width begins to increase after approximately 3 minutes. The final fracture width at 140 °C is relatively large, potentially due to the simpler fracture geometry, which results in lower flow resistance and a smaller pressure differential. Conversely, the narrower etched width at 180 °C may be attributed to the formation of low-height etched channels.

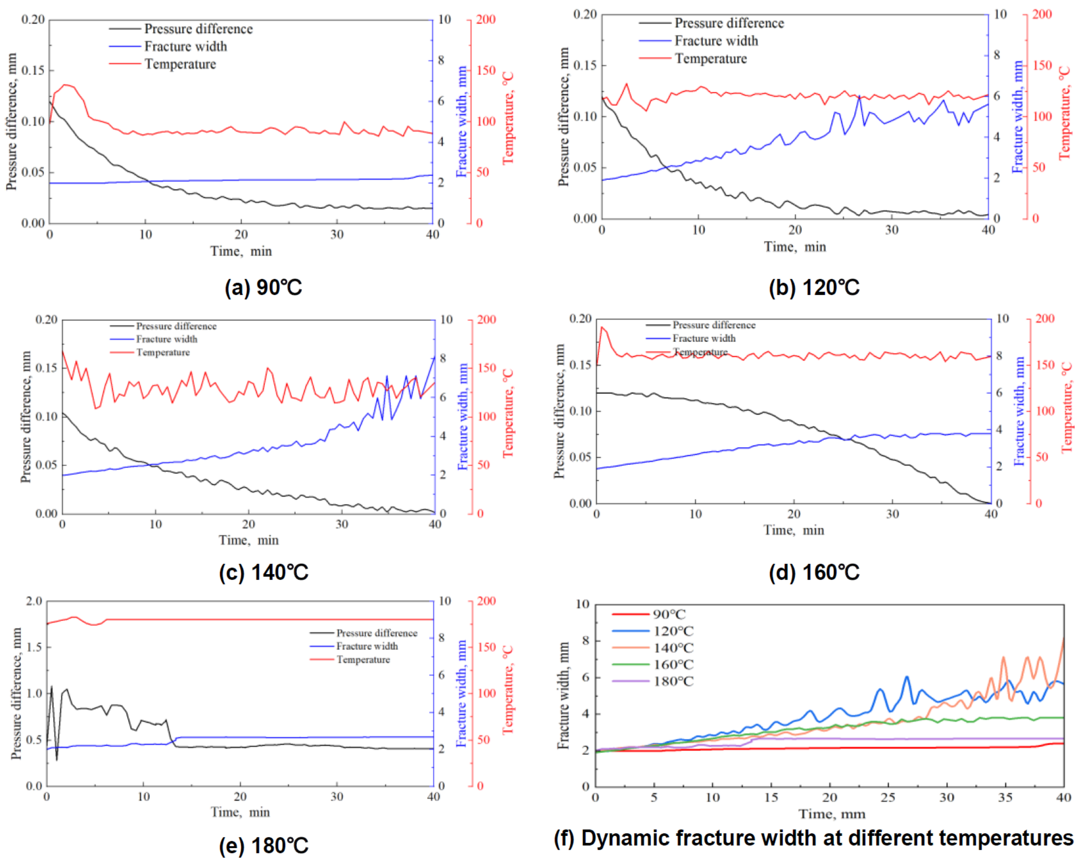

Figure 7—Pressure differential evolution and dynamic fracture during acid-etching

Acid-Rock Reaction Rate

Considering the effect of fracture surface roughness on the acid-rock reaction rate, the rate can be quantified using the following Eq. 3 (Zhang et al., 2023), which is derived from the mass of rock dissolved by the acid under varying temperature conditions:

$$J = -\frac{\partial C}{\partial t}\frac{S}{S_t} = -\frac{4\Delta m}{M_{dolomite}Vt}\frac{S}{S_t} \tag{3}$$

where, J is the reaction rate, mol/(L·s); Δm is the change in the rock sample mass before and after reaction, g; M_{dolomite} is the molar mass of dolomite, g/mol; V is the volume of acid, L; S_{t} is the true surface area of the fracture, cm^2; S is the projected surface area of the fracture, cm^2.

To investigate the effect of temperature on the acid-rock reaction rate, the rate under different temperature conditions can be predicted using the Arrhenius equation (Qi et al., 2021):

$$k = A \cdot e^{-\frac{Ea}{RT}} \tag{4}$$

$$\frac{k_2}{k_1} = e^{-\frac{E_a}{R}(\frac{1}{T_2} - \frac{1}{T_1})} \tag{5}$$

$$J = kc^2 \tag{6}$$

where, k is the reaction rate constant, s^{-1}; A is a dimensionless pre-exponential factor related to the reactive substrate; c is the concentration of H^+ in the acid, mol/L; E_a is the activation energy, J/mol; R is the universal constant, 8.314 J/(mol·K); T is the absolute temperature, K.

Fig. 8 compares the experimentally measured reaction rates with those predicted by the Arrhenius equation, with the reaction rate at 90°C serving as the reference. The experimentally data show an exponential increase in the reaction rate with temperature in the range of 90–140 °C. At 140 °C, the reaction rate is approximately 2.5 times greater than at 90 °C. When the temperature exceeds 140 °C, the reaction rate rises linearly with temperature and the reaction rate at 180 °C reaches 11.45 times the value observed at 90 °C. Moreover, the experimental results are consistently lower than the values predicted by the Arrhenius equation. This discrepancy is likely due to the limited extent of the acid-rock reaction, which results from restricted acid flow within the fracture, in contrast to the more uniform and complete reaction conditions in a fully mixed solution system.

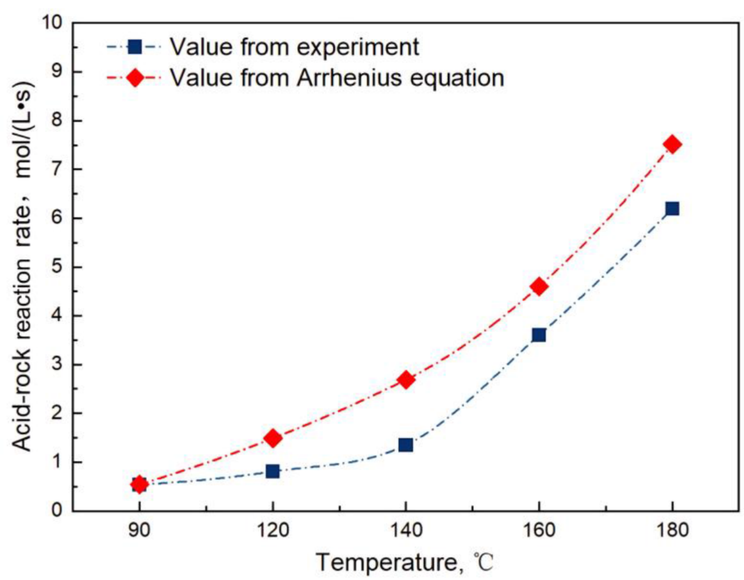

Figure 8—Acid-rock reaction rate with different temperature

Acid-Fracture Morphology

As shown in Fig. 9, at 90 °C, acid etching resulted in a slight reduction in red-colored protruded regions, a significant decrease in yellow relatively flat regions, and a slight increase in green recessed regions. These indicate that the rock reaction rate at this temperature is relatively low. When flowing through the rough fracture, acid primarily dissolves smooth surfaces and only slightly erode large protrusions. In addition, the fracture surface profile lines before and after acid-etching along the length of the rock slab (*Y* direction) reveal that acid mainly reduces the width of protrusions rather than their height.

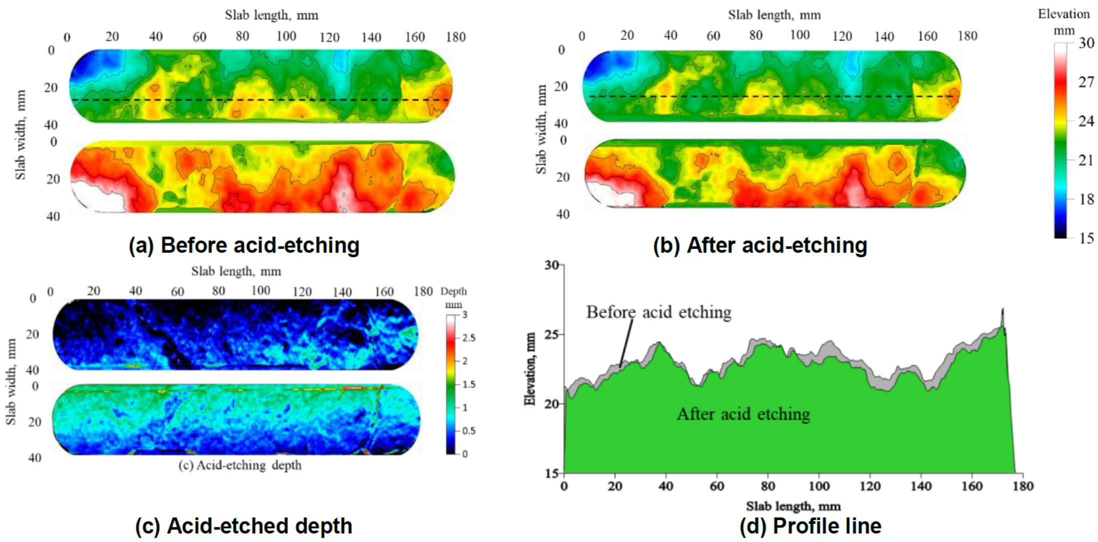

Figure 9—Elevation and etched-depth distribution of the rock plate at 90 °C

As shown in Fig. 10, the acid-etching behavior on the rough surface at 120 °C is like that observed at 90 °C; however, the dissolution depth at 120 °C is significantly greater. The profile lines indicate that, at 120 °C, acid not only reduced the width of the protrusions but also lowered their height. Additionally, acid-etched depth in large recessed areas is noticeably lower than that in the protruded regions.

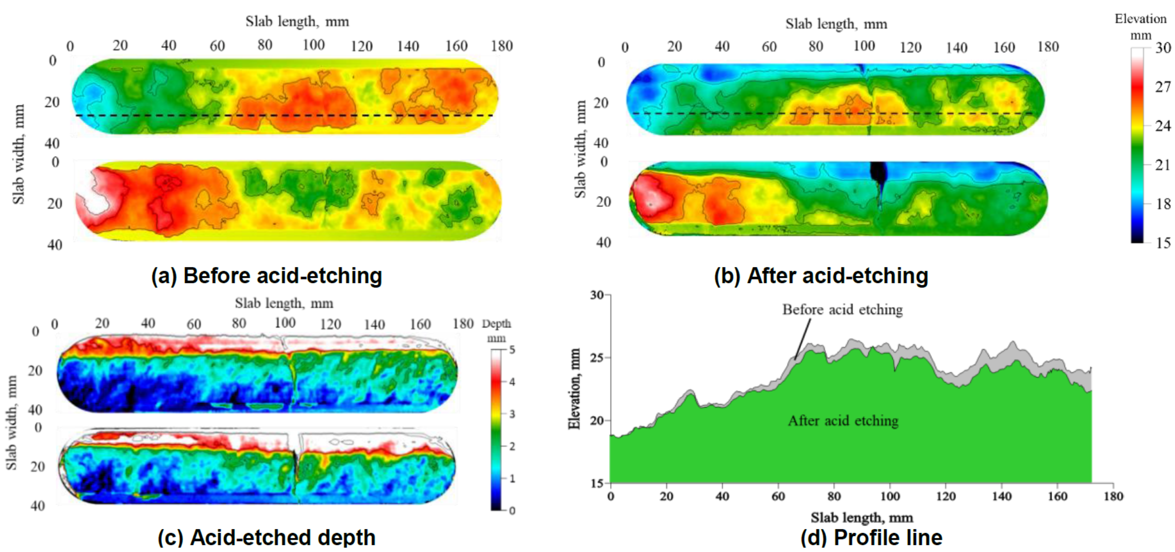

Figure 10—Elevation and etched-depth distribution of the rock plate at 120 °C

In comparison, at 140 °C, the acid dissolved a substantially greater amount of rock and the protrusions in the profile lines were almost completely flattened (Fig. 11a, b, d), indicating a more pronounced peak-removing effect. This also explains the spot-like distribution of deeper red regions in the acid etching depth map (Fig. 11c). Additionally, the green recessed area near the center of the rock slab expanded after acid etching, suggesting that the acid exhibited a valley-broadening effect at 140 °C (Fig. 11).

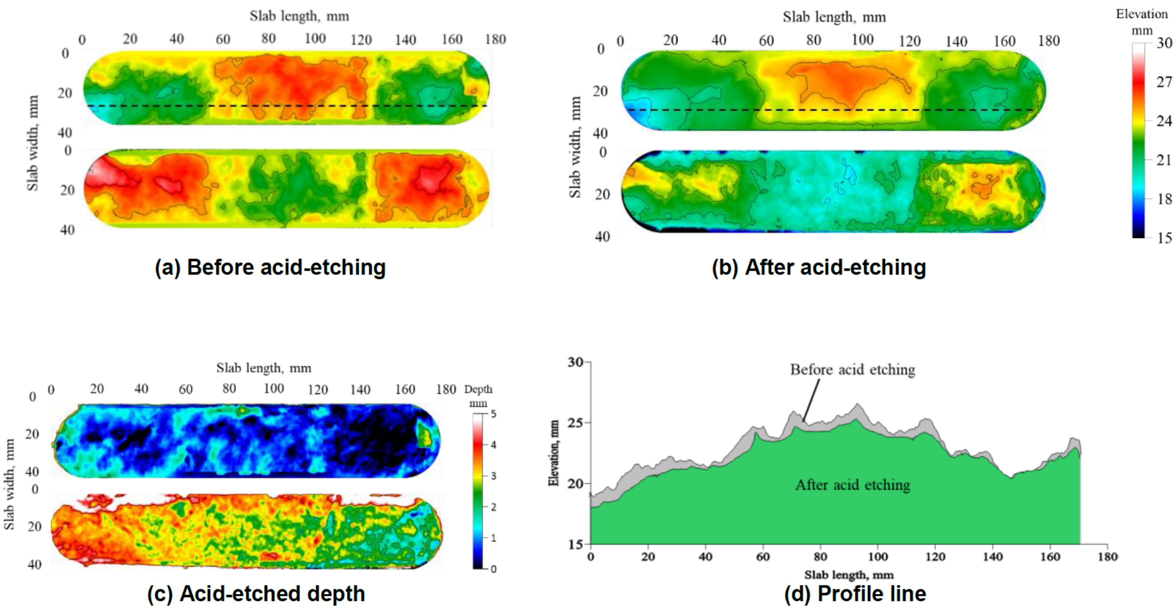

Figure 11—Elevation and etched-depth distribution of the rock plate at 140 °C

When the temperature increased to 160 °C, the proportion of white and red protruded regions on the fracture surface after acid etching significantly decreased, while the green and blue recessed areas expanded and deepened (Fig. 12a, b). This indicates that the acid at this temperature exhibited peak-removing, valley-broadening and deepening effects at the same time. The dissolution of the fracture surface became more uniform (Fig. 12c). Moreover, in the relatively flat region, acid etching resulted in the formation of burr-like protrusions (Fig. 12d). These features are responsible for the appearance of green spot-like etching pits and deep blue dot-shaped support structures. However, the acid still showed limited dissolution in large recessed areas.

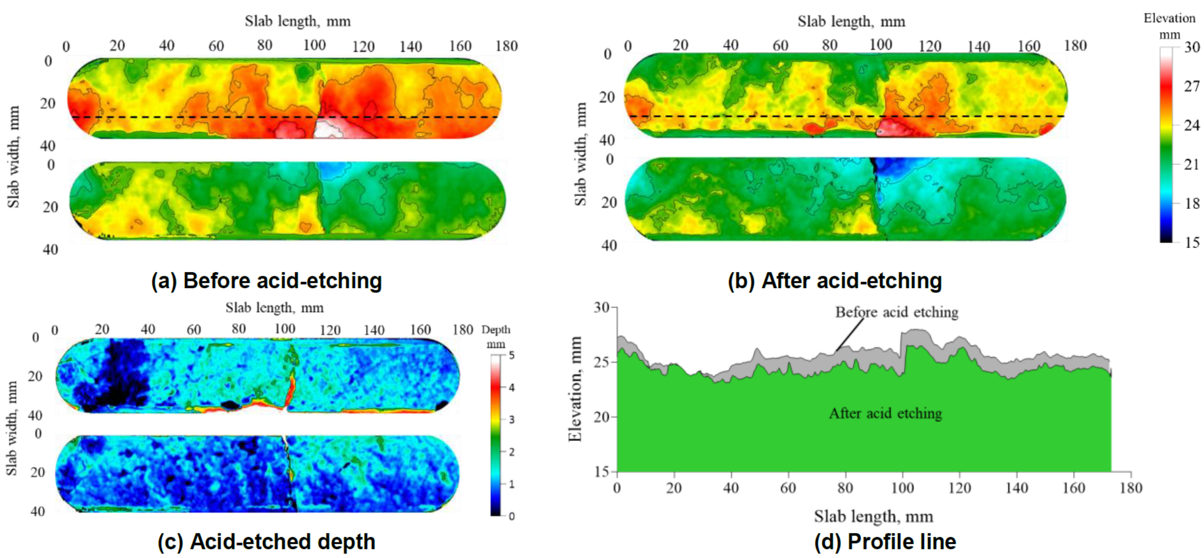

Figure 12—Elevation and etched-depth distribution of the rock plate at 160 °C

Fig. 13 demonstrates that at 180 °C, the acid exhibits similar effects of peak-removing, valley broadening and deepening on the fracture surface. The acid-etched depth reveals that acid-etching predominantly occurred on one side of the fracture, which may be attributed to the significantly great reaction rate (Fig. 13c). As a result, the acid rapidly created a flow channel and continuously enlarged it. Furthermore, the fracture surface profile confirms that the acid-etched channel is relatively uniform exhibiting minimal morphological variation along the acid-fracture (Fig. 13d).

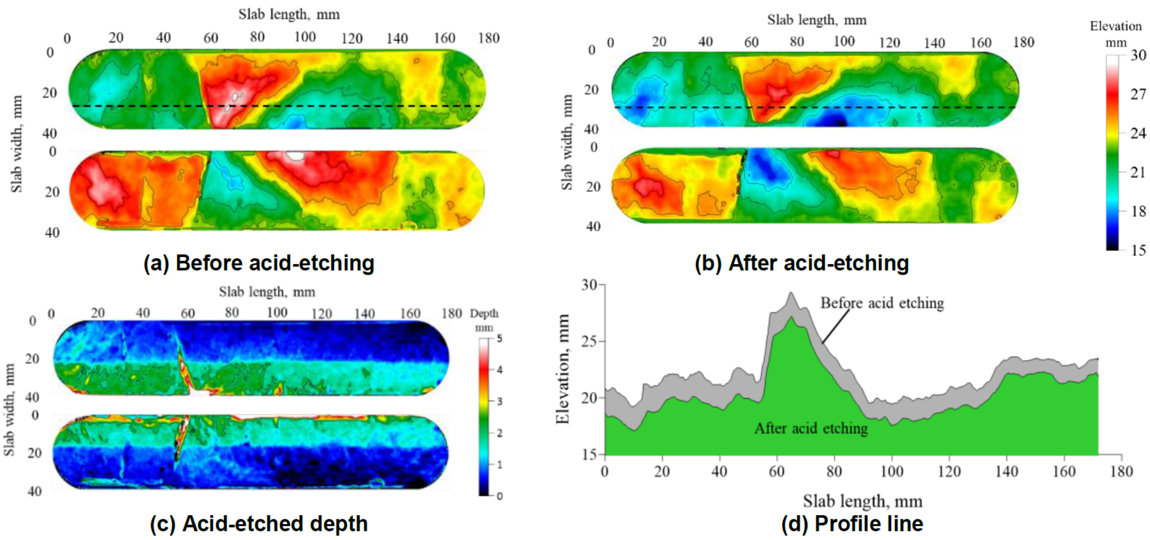

Figure 13—Elevation and etched-depth distribution of the rock plate at 180 °C

At different temperatures, the acid demonstrates distinct effects in modifying the fracture surface, including "protrusions slimming," "peaks removing," "valley widening," and "valley deepening." At 90 °C, the acid primarily causes localized dissolution on the fracture surface, predominantly leading to the slimming of protrusions, with minimal effects on valley deepening or widening. At 120 °C, the dissolution process mainly focuses on peak removing. When the temperature reaches 140 °C, little protrusions are nearly completely dissolved, and a significant valley widening becomes evident. At 160 °C, the acid continued to fully dissolve the protrusions while simultaneously promoting both valley deepening and widening. As the temperature further increases to 180 °C, extensive and uniform dissolution occurs across the entire fracture surface, resulting in the formation of smooth, unilaterally etched acid channels.

Changes in fracture surface area reflect modifications in small-scale surface morphology (Gou et al., 2021). As shown in Fig. 14,, the fracture area increases after acid-etching at 90 °C, 160 °C, and 180 °C, but decreases at 120 °C and 140 °C. Although the initial fracture area at 120 °C and 140 °C is greater than that at the other three temperatures, the acid-etched fracture area is smaller compared to the others. Furthermore, the JRC decreases after acid etching within the temperature range of 90–140 °C, increases at 160 °C, and exhibits greater fluctuations at 180 °C, although both the average and median values still decrease (Fig. 15). These phenomena may be attributed to the various acid-etching behaviors at different temperatures. At 90 °C, acid locally dissolved around protrusions but failed to change the height of the protrusions. At 140 °C, although the acid significantly dissolved proppants, the fracture surface became smoother. At 160 °C, the acid induced pronounced dissolution, forming burr-like protrusions. When the temperature reaches 180 °C, acid etching becomes more uniform, leading to a reduction in JRC. However, due to the presence of single-sided flow channels, both the fracture surface area and the range of JRC variation increase.

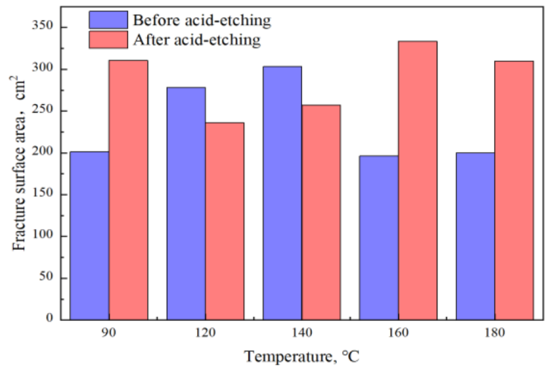

Figure 14—Variation in fracture surface area

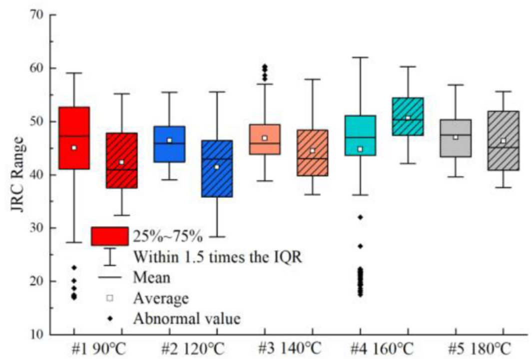

Figure 15—JRC of fracture surface

Acid-Fracture Conductivity

As shown in Fig. 16 and Fig. 17a, the initial conductivity exhibits a positive correlation with temperature. When closure pressure is below 20 MPa, conductivity decreases approximately linearly with increasing pressure at 90 °C and 120 °C. But the decrease trend becomes exponential at temperatures above 120 °C. At 20 MPa, conductivity loss at 140 °C is less than 50%, while that at 160 °C exceeds 50%. The reduction at 180 °C is smaller than at 160 °C due to the formation of a unilateral flow channel (Fig. 13c).

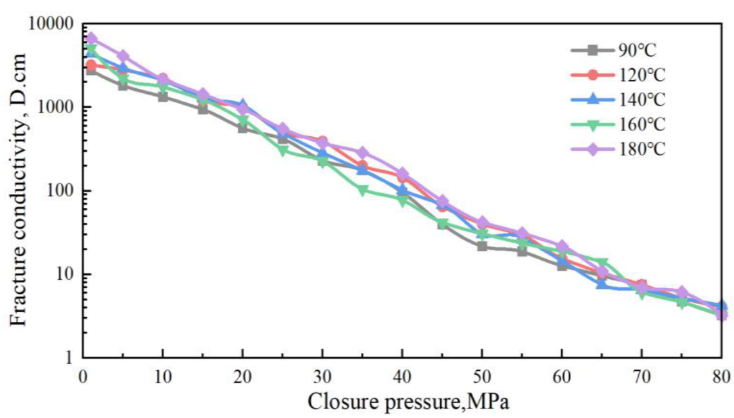

Figure 16—Conductivity of acid-fracture under closure pressure

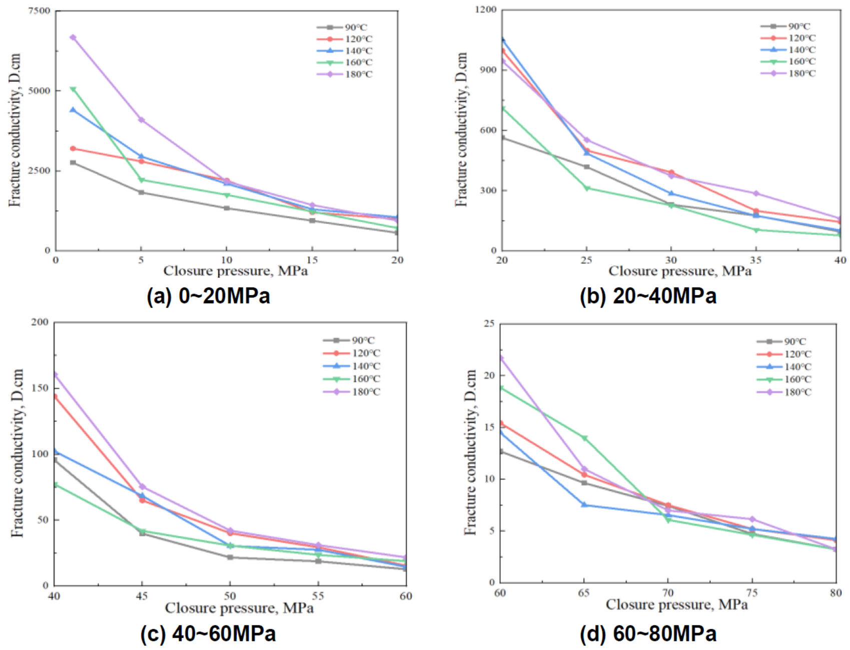

Figure 17—Acid-fracture conductivity across different closure pressure

When the closure pressure is in the range of 20-40 MPa (Fig. 17b), the conductivity at 90 °C still decreases linearly with the increasing pressure, whereas the conductivity at temperatures of 120 °C and above exhibits an exponential decline. In the range of 40-60 MPa (Fig. 17c), the conductivity decreases exponentially at all tested temperatures. When the closure pressure exceeds 60 MPa (Fig. 17d), conductivity at 90 °C decreases linearly, while temperatures above 90 °C show an exponential decline. At 80 MPa, conductivity at 180 °C is only 67% of that at 90 °C.

Initial contact fracture width calculated using the contact method (Gou et al., 2023) and the JRC of acid-fracture width are employed to evaluate their impacts on conductivity. Fig. 18 presents the initial contact width contour map, average width and JRC distribution along the slab length (*Y*-direction). At 90 °C, the dark blue regions and low JRC (Fig. 18a), indicate the narrowest fracture width and a uniform width distribution, resulting in the lowest initial acid-fracture conductivity (Fig. 17a). At this temperature, forming discrete acid-cavities (indigo blue), which correspond to a lower JRC (Fig. 18a), suggesting that the fracture morphology is dominated by small-scale cavities. At temperatures above 120 °C, JRC increases significantly, indicating more complex cavity morphology, and the fracture width widens (Fig. 18b, c, d, e), leading to higher initial conductivity (Fig. 17a). Between 120 °C and 160 °C, both narrow dark-blue regions and wider indigo-blue zones appear (Fig. 18c, d). As temperature rises, dark-blue regions gradually diminish, while discrete dark-blue support points become more distinct, increasing initial conductivity and more flow channels. At 180 °C, a clear acid-etched channel forms, leading to the highest initial conductivity. This also explains the smaller pressure differential and reduced dynamic fracture width observed during the acid-etching process (Fig. 7e).

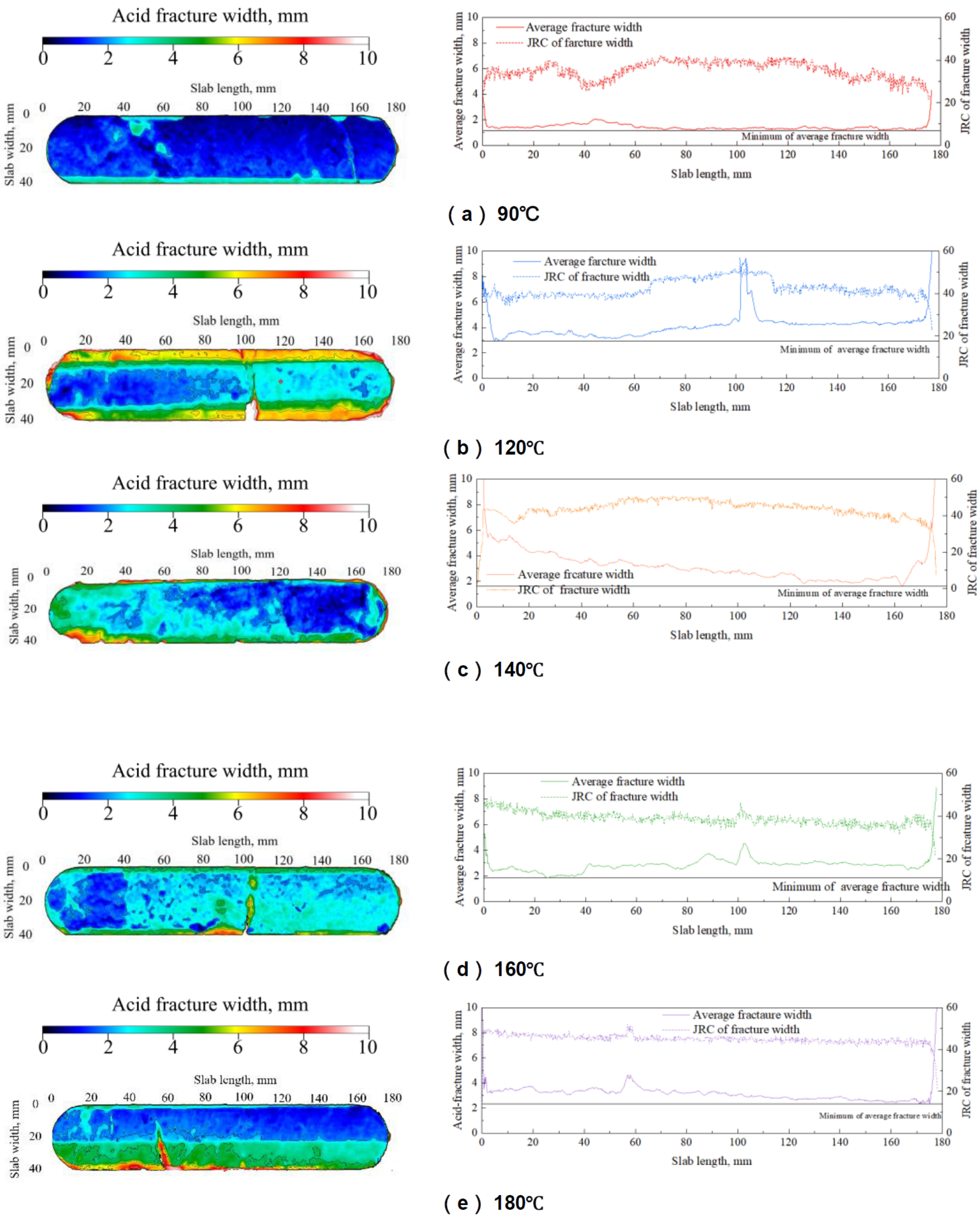

Figure 18—Average fracture width and JRC curves in the rock slab length-direction

The fracture conductivity at 80 MPa shows a clear positive correlation with the average width JRC, and it is also influenced by the average contact fracture width (Fig. 19). Between 90 °C and 160 °C, a strong relationship between conductivity and JRC is observed. However, this relationship changes until a single-sided flow channel formed at 180 °C. These observations suggest that conductivity of this kind fracture may primarily be attributed to the average fracture width, indicating that the fracture width JRC is not an appropriate parameter for this fracture morphology.

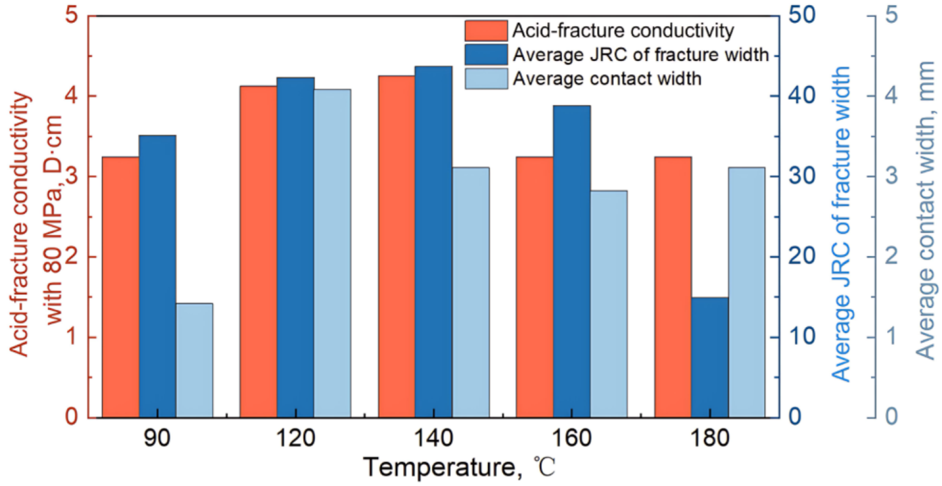

Figure 19—Fracture conductivity with 80MPa with average initial contact width and fracture width JRC

Challenges and Prospects

In experiments, the acid-rock reaction rate on the fracture surface was lower than theoretical predictions. The reaction rate used in acid etching theory-especially under ultra-high temperatures-should be based on experimental data. Although a single-sided etched channel formed at 180 °C, the fracture showed no significant conductivity under 80 MPa closure pressure. This may result from the flat channel morphology and weakened mechanical properties at high temperatures. Thus, improving conductivity while maintaining mechanical strength at ultra-high temperatures remains a key research goal for acid-etched fractures in carbonate reservoirs.

In addition, the acid aging occurred at 200 °C, as both fine sand-like solid and adsorbed clusters were observed in the acid (Fig. 20a). Simultaneously, severe corrosion of the pipelines due to failure of the corrosion inhibitor and acid leakage occurred (Fig. 20b), ultimately rendering the experiments unfeasible. These solid phases precipitated from acid may block pores and natural fractures in the near-wellbore region, forming a plugging layer that hinders acid penetrating into the deep reservoir. This significantly impedes the deep stimulation in ultra-high temperature reservoirs. If such impurities are carried into the deep reservoir by the acid, they may deposit within fractures, thereby reducing fracture conductivity. Furthermore, the severe corrosivity of acid under ultra-high temperature conditions can cause two major issues: on the one hand, it may induce localized corrosion of the tubulars, preventing the acid from reaching the target zone as intended; on the other hand, corrosion residues may further exacerbate reservoir damage (Fig. 21). Therefore, enhancing the thermal stability of acid additives, such as corrosion inhibitors and thickeners, is essential for advancing acid fracturing in ultra-high-temperature reservoirs above 200 °C.

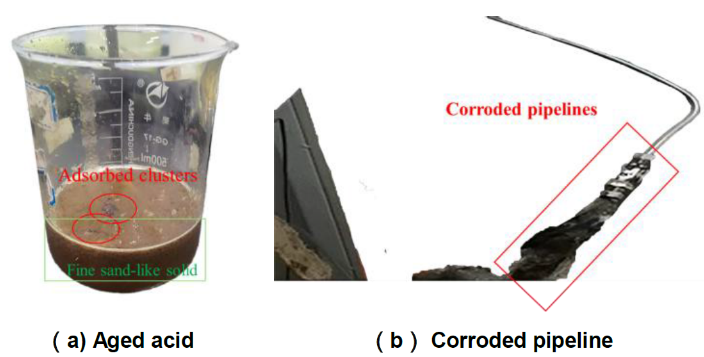

Figure 20—Challenges encountered during acid-etching experiment at 200 °C

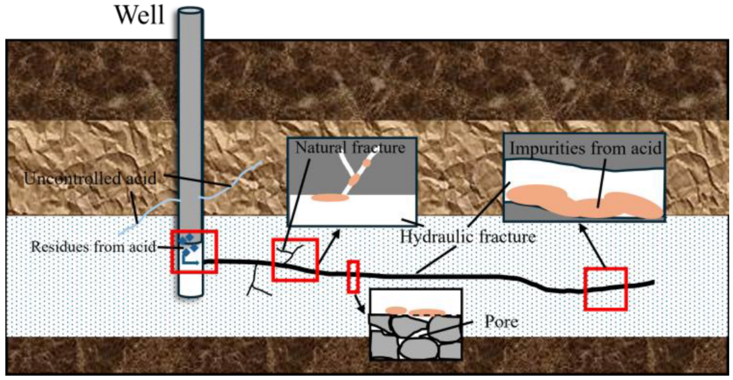

Figure 21—Damage caused by aged acid in ultra-high temperature reservoir

Conclusions

In this study, an API high-temperature fracture acid-etching device was developed to perform acid-etching experiments on fractures at temperatures ranging from 90 °C to 180 °C. The acid-etching behavior and conductivity evolution under ultra-high-temperature conditions were systematically examined in terms of pressure differential, dynamic fracture width, acid-rock reaction rate, acid-fracture morphology, initial contact width, and fracture conductivity. The main findings are summarized as follows:

- At 160 °C, both pressure differential and dynamic fracture width exhibited logarithmic growth during acid-etching, while at 180 °C they varied in stages. In contrast, between 90-140 °C, the pressure differential decreased exponentially, and the dynamic fracture width increased linear-logarithmic.
- Between 90°C and 140 °C, the acid-rock reaction rate increased exponentially, reaching 2.5 times the 90 °C level at 140 °C. Above 140 °C, the increase became linear, reaching 11.45 times the 90°C rate by 180 °C.
- At 160 °C, acid uniformly dissolved surface protrusions while widening and deepening valleys, resulting in a relatively uniform width distribution. At 180 °C, acid etched a single-sided flow channel. Below 140 °C, the peak-removing and valley-widening effects of acid were weaker but strengthened with rising temperature, leading to greater variability in initial contact width.
- Fracture conductivity increased proportionally with temperature and was not solely determined by initial contact width. At 140 °C and above, conductivity declined exponentially with closure pressure, with slower decline rate at higher rises. Below 140 °C, the decline followed a linear-exponential trend. Under 80 MPa, conductivity at or below 160 °C correlated positively with the complexity of initial width distribution. Although acid formed a single-sided channel at 180 °C that enhanced conductivity, final conductivity was only 67% of that at 90 °C.

Acknowledgments

The authors appreciate the financial support for this research from Joint Fund Project for Science and Technology & Education of Sichuan Province (No. 2024NSFSC2004) and National Natural Science Foundation of China (No. U21A20105).

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